INCH-POUND

MIL-C-83513/4E
31 March 1989
SUPERSEDING
MIL-C-83513/4D
7 March 1986

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, SOCKET CONTACTS, CLASS M, CRIMP TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83513.

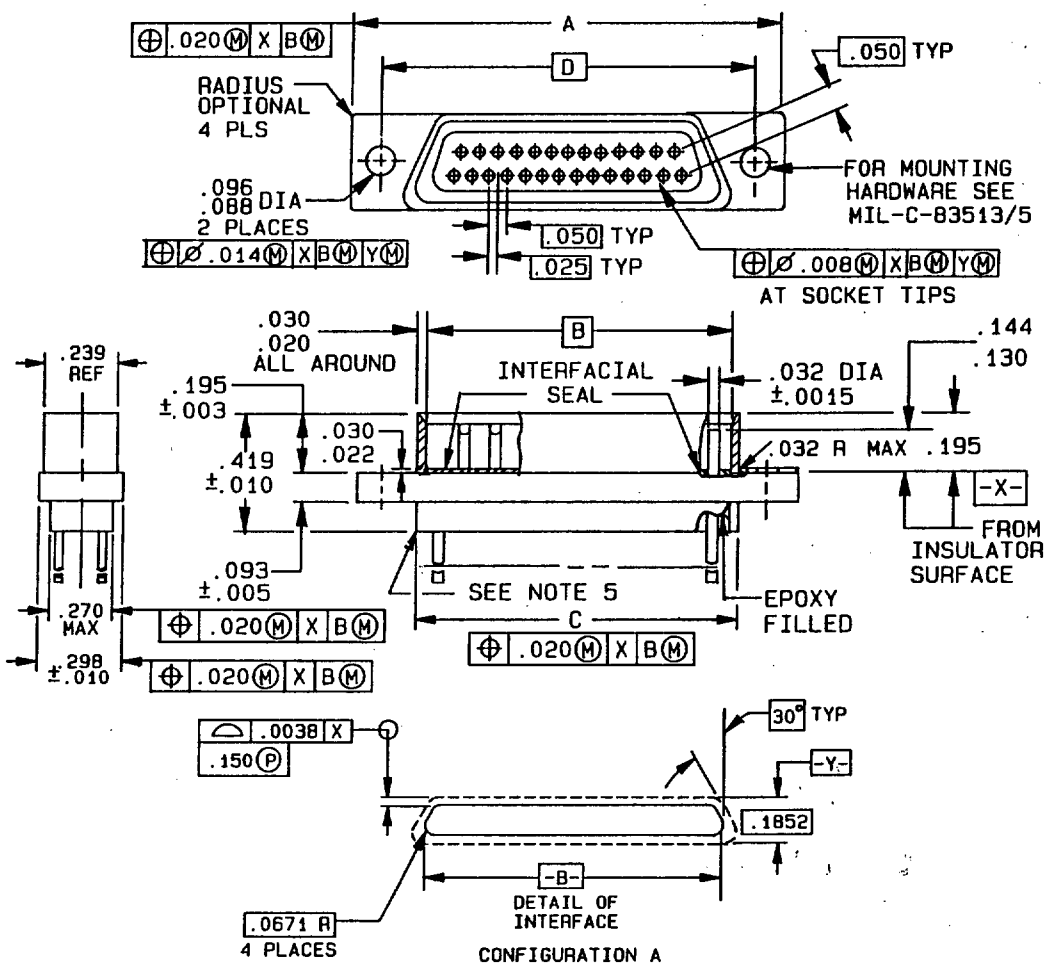


FIGURE 1. Connector, receptacle (.050 spacing).

Ⓔ denotes change

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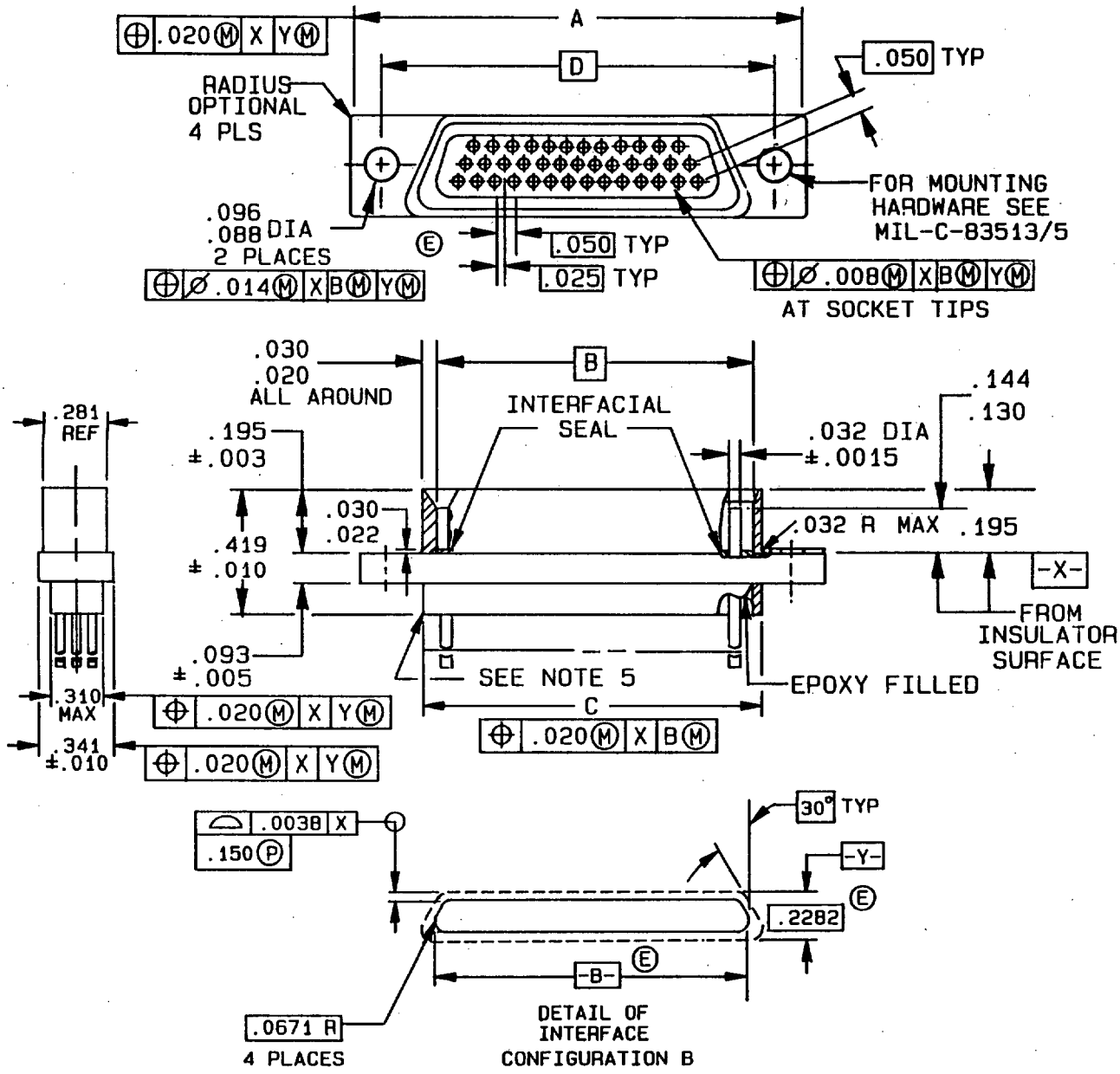


FIGURE 1. Connector, receptacle (.050 spacing) - Continued.

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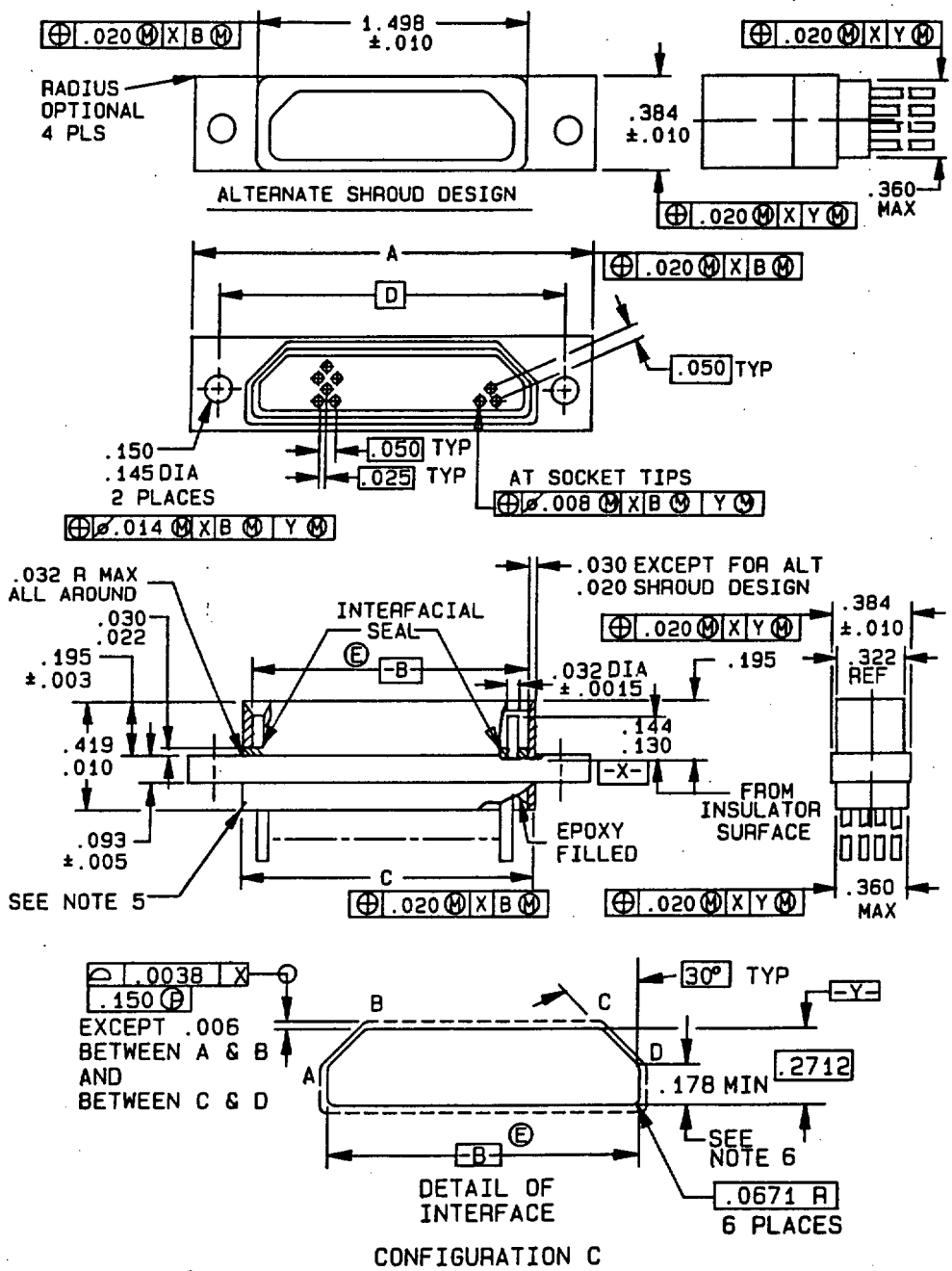


FIGURE 1. Connector, receptacle (.050 spacing) - Continued.

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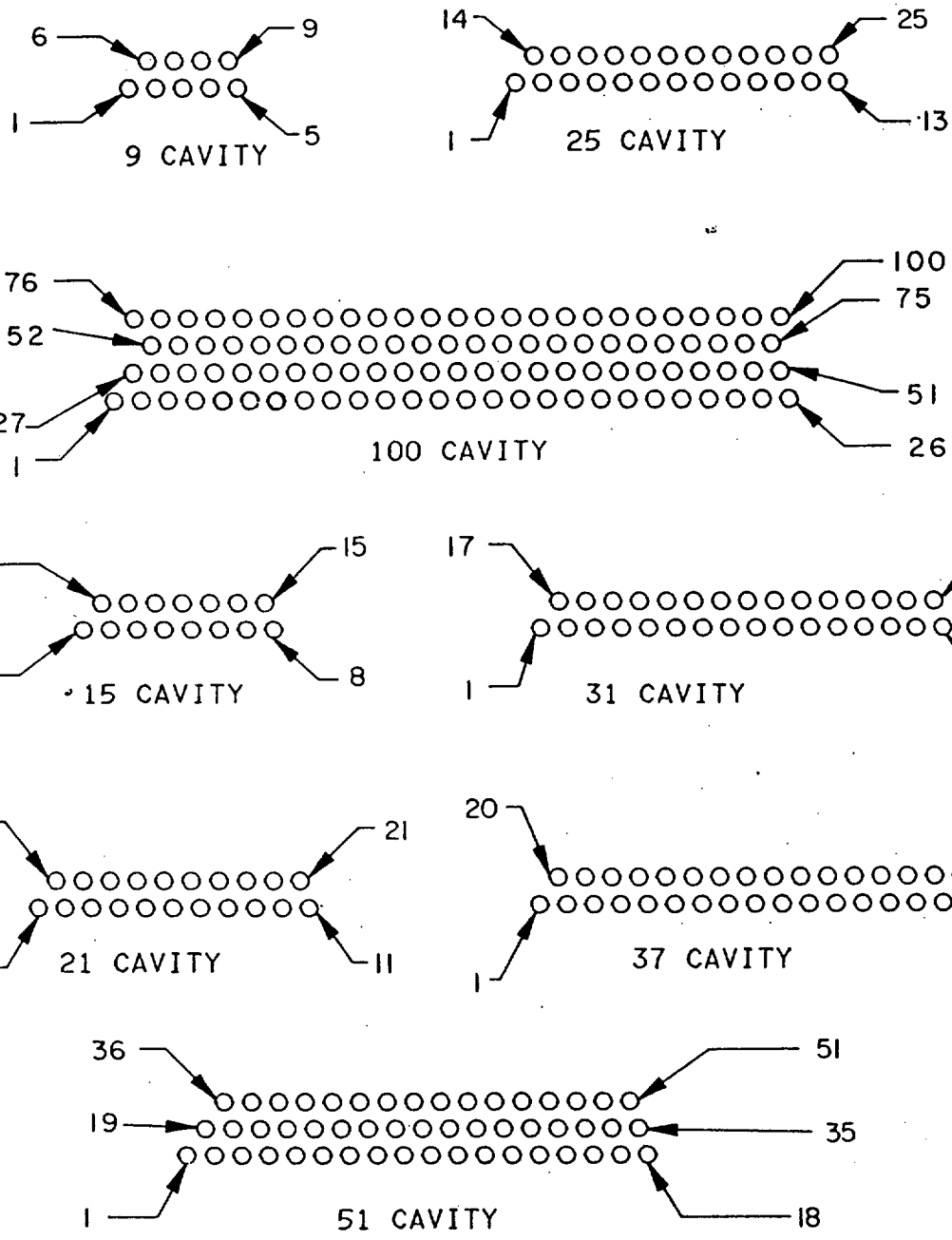
Number cavities	A ±.010 (.25)	B BSC	C ±.010 -.018 +(.25) -(.46)	D BSC	Config. letter
9	.775 (19.68)	.3342 (8.489)	.390 (9.91)	.565 (14.35)	A
15	.925 (23.50)	.4842 (12.299)	.540 (13.72)	.715 (18.16)	A
21	1.075 (27.30)	.6342 (16.109)	.690 (17.53)	.865 (21.97)	A
25	1.175 (29.84)	.7342 (18.649)	.790 (20.07)	.965 (24.51)	A
31	1.325 (33.66)	.8842 (22.459)	.940 (23.88)	1.115 (28.32)	A
37	1.475 (37.46)	1.0342 (26.269)	1.090 (27.69)	1.265 (32.13)	A
51	1.425 (36.20)	.9842 (25.00)	1.040 (26.42)	1.215 (30.86)	B
100	2.160 (54.86)	1.3842 (35.159)	1.432 (36.37)	1.800 (45.72)	C

Inches	mm	Inches	mm
.002	0.05	.096	2.44
.003	0.08	.100	2.54
.0035	0.089	.1141	2.898
.0038	0.097	.117	2.97
.004	0.10	.123	3.12
.005	0.13	.130	3.30
.006	0.15	.144	3.66
.007	0.18	.145	3.68
.008	0.20	.150	3.81
.010	0.25	.180	4.57
.014	0.36	.1852	4.704
.018	0.46	.189	4.80
.020	0.51	.193	4.90
.022	0.56	.195	4.95
.023	0.58	.200	5.08
.025	0.64	.2282	5.796
.030	0.76	.232	5.89
.032	0.81	.239	6.07
.033	0.84	.270	6.86
.045	1.14	.2712	6.888
.050	1.27	.275	6.99
.067	1.702	.281	7.14
.0671	1.704	.298	7.57
.069	1.75	.310	7.87
.070	1.78	.322	8.18
.088	2.24	.341	8.66
.090	2.29	.360	9.14
.093	2.36	.384	9.75
.094	2.39	.419	10.64
.095	2.41	1.438	36.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is .005 (0.13 mm).
4. Metric equivalents are in parentheses or columns.
5. Point from which lead length is measured.
6. Dimension line at point D locates the point of intersection of the two adjacent sides of the receptacle interface.

FIGURE 1. Connector, receptacle (.050 spacing) - Continued.

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NOTE: Engaging face of socket insert shown.
Cavity identification numbers are for reference only and do not appear on part.

FIGURE 2. Insert arrangement.

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REQUIREMENTS:

Dimensions and configuration: See figure 1. This specification sheet describes the socket side of a rectangular metal shell connector. This connector uses reverse gender contacts, i.e., the live pin recessed in the insulator with the static socket protruding from a shrouded interface.

Material and finish:
Interfacial seal: Silicone rubber or fluorosilicone rubber or blend thereof.

Contact identification: See figure 2 for contact layout numbering.

Contact connection: The pin contact (being recessed in the insulator) is normally connected to the live side of the circuit.

Current rating, maximum: 3 amperes per contact.

Mounting and mating hardware: Mounting hardware is to be ordered separately. For insert arrangements A through G, use MIL-C-83513/5, configurations A and B. For insert arrangement H, use MIL-C-83513/5, configuration C (see figure 2).

Mating plug: Shall conform to MIL-C-83513/1 or MIL-C-83513/3.

Ⓔ Part number: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled from the code.

M83513/04- A 01 1/ C

- M83513/04-: Specification sheet number
- A: Insert arrangement (see fig. 2)
  - A = 9
  - B = 15
  - C = 21
  - D = 25
  - E = 31
  - F = 37
  - G = 51
  - H = 100
- 01: Wire type
  - 01 = M22759/11-26-9, 18 inches long 2/ 3/
  - 02 = M22759/11-26-9, 36 inches long 2/ 3/
  - 03 = M22759/11-26-( ), 18 inches long 3/ 4/ 5/
  - 04 = M22759/11-26-( ), 36 inches long 3/ 4/ 5/
  - 05 = QQ-W-343, type S, 25 AWG, except shall be gold plated 10 microinches; 0.5 inch long 6/
  - 06 = QQ-W-343, type S, 25 AWG, except shall be gold plated 10 microinches; 1.0 inch long 6/
  - 07 = QQ-W-343, type S, 25 AWG, tin plated 100 microinches; 0.5 inch long 6/
  - 08 = QQ-W-343, type S, 25 AWG, tin plated 100 microinches; 1.0 inch long 6/
  - 09 = M22759/33-26-9, 18 inches long 3/
  - 10 = M22759/33-26-9, 36 inches long 3/
  - 11 = M22759/33-26-( ), 18 inches long 3/ 4/
  - 12 = M22759/33-26-( ), 36 inches long 3/ 4/
  - 13 = M22759/11-26-9, 72 inches long 3/
  - 14 = M22759/11-26-( ), 72 inches long 3/
  - 15 = M22759/33-26-9, 72 inches long 3/
  - 16 = M22759/33-26-( ), 72 inches long 3/
- C: Shell finish
  - C = Cadmium
  - N = Electroless nickel (Space applications only)

See footnotes on next page.

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1/ These connectors have leads attached of which wire type, wire size, wire color, and lead length is specified in part number.
2/ For connectors with shell finish N, wire shall be M22759/33-26-9.
3/ Lead length tolerance +1.0, -0.0 inch.
4/ Color coding in accordance with MIL-STD-681, system I, except the appropriate color code shall be the individual contact cavity number minus 1 for contact positions 1 through 10, 11 through 70, etc. For example:

Pin number	Identification number
4	3
9	8
21	0
31	0
77	6
100	9

This color coding system is to be implemented within 12 months after the date of this document.

5/ For connectors with shell finish N, wire shall be M22759/19-26-( ).
6/ Lead length tolerance +0.2, -0.0 inch.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3637-04)

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